Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int freq[n + 1];
    for (int i = 0; i <= n; i++) {
        freq[i] = 0;
    }
    for (int i = 0; i < n; i++) {
        int x;
        scanf("%d", &x);
        freq[x]++;
        if (freq[x] == 2) {
            printf("%d\n", x);
            return 0;
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int T;
    scanf("%d", &T);
    while (T--) {
        int n1, n2;
        scanf("%d", &n1);
        int a[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }
        scanf("%d", &n2);
        int b[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }
        int i = 0, j = 0;
        int first = 1;
        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                if (first || a[i] != a[i - 1]) {
                    if (!first) {
                        printf(" ");
                    }
                    printf("%d", a[i]);
                    first = 0;
                }
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    int freq[n + 1];
    for (int i = 0; i <= n; i++) {
        freq[i] = 0;
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    for (int i = 0; i < n; i++) {
        freq[a[i]]++;

        if (freq[a[i]] == 2) {
            printf("%d\n", a[i]);
            return 0;
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n, k;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    scanf("%d", &k);
    int i = 0, j = 1;
    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }
        int diff = a[j] - a[i];
        if (diff == k) {
            printf("1\n");
            return 0;
        }
        else if (diff < k) {
            j++;
        }
        else {
            i++;
        }
    }
    printf("0\n");
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
int main() {
    int n, k;
    scanf("%d", &n);
    int *a = malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    scanf("%d", &k);
    int i = 0, j = 1;
    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }
        long long diff = (long long)a[j] - a[i];
        if (diff == k) {
            printf("1\n");
            free(a);
            return 0;
        }
        else if (diff < k) {
            j++;
        }
        else {
            i++;
        }
    }
    printf("0\n");
    free(a);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.